MariltonCR Conjecture on Nested Primes

Each of the 9 blocks contains an m-rough number for $m \geq 10^6$

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Abstract

We show that, for every $m \geq 10^6$ and each $x \in \{0, \ldots, 8\}$, the block $K_x = \{m^2 + xm + 1, \ldots, m^2 + (x+1)m\}$ contains at least one m-rough integer ($P^-(n) > m$). The proof combines Buchstab's identity and function ω with the explicit Mertens formula via Abel summation, yielding a uniform local lemma on short intervals $[X, X+Y]$ with $X \in [y^2, y^3]$ and $y \leq Y \leq 2y$:

$$\#\{X < n \leq X + Y : \; P^-(n) > y\} \;\geq\; \frac{Y}{\log y}\,\omega\Big(\frac{\log X}{\log y}\Big) \;-\; C_1 \frac{Y}{\log^2 y} \;-\; C_2,$$

with effective constants $C_0 \leq 2$, $C_1 \leq 4.4$, and $C_2 = 100$. We treat the boundary $X = y^2$ separately and remove endpoint terms by the convention $\omega(u) = 0$ for $0 < u < 1$. Applying the local lemma with $y = Y = m$ and $X = m^2 + xm$, we obtain the result for each K_x. With the current constants, numerical calculations suggest that the same argument would already work from $m \geq 18\,794$; however, in this work we formally certify $m \geq 10^6$ and cover $2 \leq m < 10^6$ via a reproducible computational sweep (see Remark 7.1).

In Lean, the finite range $2 \leq m < 10^6$ is imported as a single external axiom backed by a deterministic sweep (Zenodo doi.org/10.5281/zenodo.17137291), while the asymptotic bridge for $m \geq 10^6$ is validated uniformly by a Julia interval certificate available in the project's code repository (github.com/mariltondev/rough-blocks-lean; file `certs/uniformbridgege1e6.json`, generated by `external/julia/uniform_bridge_certificate.jl`).

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Contents

1 Introduction

We study m-rough integers (that is, $P^{-}(n) > m$) in short windows near m^2. This repository isolates the technical core that underpins a broader program — hereafter nicknamed "Conjectura de MariltonCR dos primos encaixados". The present work is entirely *unconditional*: we prove a uniform local result guaranteeing, for large m, the presence of an m-rough integer in each of nine consecutive blocks of length m immediately to the right of m^2.

In parallel papers, we develop the combinatorial and sieve implications of this framework to settle some classical conjectures: **Legendre's Hypothesis**, **Oppermann's Conjecture**, **Brocard's Conjecture** (in the form "≥ 4 primes between n^2 and $(n+1)^2$"), and the **strong Goldbach Conjecture** (conditional on strengthened versions of the program), as well as rederiving **Bertrand's Postulate** as a corollary. In parallel, we open a line of study connecting these techniques to sieve heuristics under the **Generalized Riemann Hypothesis (GRH)**.

These goals are deliberately ambitious: we already present them as a research roadmap, and there are links to this article. Here we deliver the first technical brick — a local theorem with explicit constants — upon which the subsequent stages are being developed in separate preprints, with periodic updates.

Main result. For every $m \geq 10^6$ and every $x \in \{0, \ldots, 8\}$, the block

$$K_x = \{\, m^2 + xm + 1, \ldots, m^2 + (x+1)m \,\}$$

contains at least one m-rough integer (Theorem 6.1). With the explicit constants we obtain, the same argument already works for $m \geq 18{,}794$ (Remark 6.2); we keep the statement with $m \geq 10^6$ for conservatism and to emphasize the margin.

Idea of the proof. The proof combines: (i) Buchstab's identity and the global lemma for $\Phi(x, y)$ with error $O(x/\log^2 y)$ (Lemma 3.1); (ii) the explicit form of Mertens handled via Abel; and (iii) a fine control of the variation of Buchstab's function ω on $u \in [1, 3]$ (Lemmas 3.2 and 3.3). The central technical piece is an *aggregated short increment* (Lemma 4.1) valid for $X \in (y^2, y^3]$ and $y \leq Y \leq 2y$, while the boundary $X = y^2$ is handled separately (Lemma 4.3). Endpoint terms arising after Abel are removed by adopting the standard convention $\omega(u) = 0$ for $0 < u < 1$. The constants involved ($C_0 \leq 2,\ C_1 \leq 4.4,\ C_2 = 100$) are effective.

Formalization in *Lean 4*.

Theorem 6.1 is formalized in the repository:

$$\boxed{\textbf{github.com/mariltondev/rough-blocks-lean}}$$

as `RoughBlocks.Light.main_theorem_ge_1e6`, with statement $\forall\, m, x \in \mathbb{N},\ m \geq 10^6 \wedge x \leq 8 \Rightarrow \mathrm{LB}(m, x) \geq 1$.

The axiom audit in `RoughBlocks/Light/Spec.lean` shows two external, reproducible dependencies: (i) the finite range $2 \leq m < 10^6$, encapsulated by a single axiom `RoughBlocks.Light.budget_verified_by_sc` and backed by a deterministic sweep (Zenodo doi.org/10.5281/zenodo.17137291); and (ii) the uniform bridge for $m \geq 10^6$, given by `RoughBlocks.External.bridge_ge_1e6_uniform`, instantiated from a Julia certificate available in the code repository (github.com/mariltondev/rough-blocks-lean; `certs/uniformbridgege1e6.json`, generated by `external/julia/uniform_bridge_certificate.jl`). Everything else is proved internally in Lean (only the system's classical axioms).

Organization. Section 2 fixes definitions and conventions. Section 3 collects the base inputs for Buchstab and explicit Mertens. The aggregated short increment is established in Section 4 and the local lemma with uniform error in Section 5. Theorem 6.1 and Remark 6.2 appear in Section 6. Appendix A details the proof of $C_0 \leq 2$ and the short-error lemmas used in the body.

2 Definitions and conventions

For $n \geq 1$, let $P^-(n)$ denote the least prime dividing n (and set $P^-(1) := +\infty$). We say that n is *m-rough* if $P^-(n) > m$.

Buchstab's function $\omega(u)$, $u > 1$, is the piecewise continuous solution of

$$\omega(u) = \frac{1}{u} \ (1 \leq u \leq 2), \qquad (u\,\omega(u))' = \omega(u-1) \ (u > 2).$$

Equivalently, and often more convenient:

$$u\,\omega'(u) + \omega(u) = \omega(u-1) \ (u > 2).$$

Conventions. We use:

(C1) log denotes the natural logarithm;

(C2) we extend $\omega(u)$ by zero on $(0, 1)$, i.e., $\omega(u) = 0$ for $0 < u < 1$ and $\omega(1) = 1$. When applying the *Abel summation combined with the explicit Mertens form*, we choose the upper limit so that the argument of ω there is < 1 (e.g., $t = \sqrt{X+Y}$), killing the endpoint term without changing the classical value at $u = 1$.

This extension is standard in the Buchstab literature and allows sums up to X even when $\frac{\log(X/p)}{\log p} \leq 1$ (those terms vanish).

3 Base results

Buchstab global lemma

Lemma 3.1 (Global)**.** ***Fix*** *$U_0 := 10^3$. There exists an absolute effective constant $C_0 > 0$ such that, for $y \geq U_0$, $x \geq y$ and $u = \frac{\log x}{\log y} \in [1, 3]$,*

$$\Phi(x, y) \;:=\; \#\{n \leq x : \; P^-(n) > y\} \;=\; \frac{x}{\log y}\,\omega(u) \;+\; E(x, y), \qquad |E(x, y)| \;\leq\; C_0\,\frac{x}{\log^2 y}.$$

Reference: *Halberstam–Richert,* Sieve Methods*, Thm. III.6.4 (Ch. III §6, after Thm. III.6.4); see also Montgomery–Vaughan, Ch. 7. In Appendix A we give an explicit proof that $C_0 \leq 2$ for $y \geq U_0{=}10^3$; in any case $m \geq 10^6 \gg U_0$.*

Fine control for ω on $[1, 2]$ and $[2, 3]$

Lemma 3.2 (Exact variation for $1 < u \leq 2$)**.** *For $1 < u \leq 2$ one has $\omega(u) = 1/u$, hence $\omega'(u) = -1/u^2$. In particular, for $\delta \geq 0$,*

$$\omega(u + \delta) \;\geq\; \omega(u) \;-\; \delta \qquad \textit{whenever } u, u{+}\delta \in (1, 2].$$

Reference: *follows from the definition of ω and Buchstab's ODE on $[1, 2]$.*

Lemma 3.3 (Closed form and variation for $2 \leq u \leq 3$)**.** *For $2 \leq u \leq 3$ one has the closed expression*

$$\omega(u) \;=\; \frac{1 + \log(u - 1)}{u}.$$

In particular,

$$\omega'(u) \;=\; \frac{\frac{u}{u-1} - \bigl(1 + \log(u - 1)\bigr)}{u^2} \qquad (2 < u < 3),$$

and the uniform bound

$$\sup_{2 \leq u \leq 3} |\omega'(u)| \;\leq\; \frac{1}{4}.$$

Thus, for any $\delta \geq 0$ with $u, u{+}\delta \in [2, 3]$,

$$\omega(u + \delta) \;\geq\; \omega(u) \;-\; \frac{\delta}{4}.$$

Reference: *obtained by integrating $(u\,\omega(u))' = \omega(u - 1)$ with condition $\omega(u) = 1/u$ on $[1, 2]$; see, e.g., Montgomery–Vaughan, Ch. 7.*

Remark 3.4 (We do not use monotonicity of ω on $[2, 3]$). The derivative $\omega'(u)$ may change sign on $[2, 3]$ (e.g., positive at $u = 2$ and negative later); in local estimates, we use only the uniform control $\sup_{2 \leq u \leq 3} |\omega'(u)| \leq \frac{1}{4}$, without any appeal to monotonicity. This ensures all variation bounds for ω depend only on $|u_2 - u_1|$.

Auxiliary Mertens sum on $[y^2, y^3]$

Lemma 3.5 (Partial sum)**.** *If $y \ge U_0$ and $X \in [y^2, y^3]$, then*

$$\sum_{y<p\le X} \frac{1}{p\log p} \;=\; \frac{1}{\log y} \;-\; \frac{1}{\log X} \;+\; O\Big(\frac{1}{\log^2 y}\Big),$$

with an absolute effective constant.

Proof. Apply *Abel summation combined with the explicit Mertens form* to $A(t) = \sum_{p\le t} \frac{1}{p}$; with the effective Mertens form $A(t) = \log\log t + B_1 + R_1(t)$ and $|R_1(t)| \le A/\log t + B/\log^2 t$ for $t \ge 10^3$ (*Rosser–Schoenfeld, Thm. 20, Eq. (4.7)*; *Dusart, Thm. 6.10*). Since $X \in [y^2, y^3]$, evaluation of the resulting integral yields $\sum_{y<p\le X} \frac{1}{p\log p} = \frac{1}{\log y} - \frac{1}{\log X} + O(1/\log^2 y)$ uniformly in X on that range. In particular, for $t \ge 10^3$ one may take $A \le 1$ and $B \le 1$. $\square$

4 Short increment for Φ — aggregated version

Lemma 4.1 (Short increment — aggregated version). *If $y \geq U_0$, $X \in (y^2, y^3]$ and Y with $y \leq Y \leq 2y$, writing $u := \dfrac{\log X}{\log y}$, then*

$$\sum_{y<p\leq\sqrt{X+Y}} \Big(\Phi\Big(\frac{X+Y}{p}, p\Big) - \Phi\Big(\frac{X}{p}, p\Big) \Big) \;=\; \frac{Y}{\log y}\,\omega(u) \;+\; O\Big(\frac{Y}{\log^2 y}\Big),$$

uniformly under the hypotheses.

Proof. For $y < p \leq \sqrt{X+Y}$, write

$$\Phi\Big(\frac{t}{p}, p\Big) \;=\; \frac{t/p}{\log p}\,\omega\Big(\frac{\log(t/p)}{\log p}\Big) \;+\; E\Big(\frac{t}{p}, p\Big) \qquad (t \in \{X, X+Y\}),$$

i.e., take E as the residue after subtracting the main term (which vanishes when $\frac{\log(t/p)}{\log p} < 1$ by the convention $\omega = 0$ on $(0,1)$). When $\frac{\log(t/p)}{\log p} \in [1,3]$ we apply Lemma 3.1 to bound $|E|$; in cases with argument < 1, we do not invoke Lemma 3.1 and control only the *increment* of E via Lemma A.3.

Taking the difference at $t = X+Y$ and $t = X$ and summing over $y < p \leq \sqrt{X+Y}$, write $S_{\mathrm{tot}} = M_1 + M_2 + R$, where

$$M_1 := \sum_{y<p\leq\sqrt{X+Y}} \frac{Y/p}{\log p}\,\omega\Big(\frac{\log(X/p)}{\log p}\Big), \qquad M_2 := \sum_{y<p\leq\sqrt{X+Y}} \frac{X+Y}{p\,\log p}\Big[\omega\Big(\tfrac{\log((X+Y)/p)}{\log p}\Big) - \omega\Big(\tfrac{\log(X/p)}{\log p}\Big)\Big],$$

and $R := \sum_{y<p\leq\sqrt{X+Y}} \big(E(\frac{X+Y}{p}, p) - E(\frac{X}{p}, p)\big)$.
Technical detail (Abel $\Rightarrow$ main term ω). With $y \geq U_0$, let $A_y(t) := \sum_{y<p\leq t} 1/p$ and $g(t) := \dfrac{\log y}{\log t}\,\omega\big(u(t)\big)$, where $u(t) := \frac{\log(X/t)}{\log t}$. The sum

$$M_1 = \sum_{y<p\leq\sqrt{X+Y}} \frac{Y/p}{\log p}\,\omega\Big(\frac{\log(X/p)}{\log p}\Big)$$

satisfies, by Abel summation,

$$M_1 = \frac{Y}{\log y}\Big[A_y(t) g(t)\Big]_{t=y}^{t=\sqrt{X+Y}} - \frac{Y}{\log y}\int_y^{\sqrt{X+Y}} A_y(t)\, g'(t)\, dt.$$

The boundary terms vanish because $A_y(y) = 0$ and $\frac{\log(X/\sqrt{X+Y})}{\log\sqrt{X+Y}} < 1$ (hence $\omega = 0$ by convention). Differentiating g and using $(u\,\omega(u))' = \omega(u-1)$, together with $A_y(t) = \log\log t - \log\log y + O(1/\log y)$ (explicit Mertens, uniform for $t \geq y \geq U_0$), we obtain

$$M_1 = \frac{Y}{\log y}\,\omega\Big(\frac{\log X}{\log y}\Big) \;+\; O\Big(\frac{Y}{\log^2 y}\Big) \quad \text{(with no boundary term).}$$

Treating M_2. Split $M_2 = M_{2,1} + M_{2,2}$ with

$$M_{2,1} := \sum_{y<p\leq\sqrt{X}} \frac{X+Y}{p\,\log p}\Big[\omega\Big(\frac{\log((X+Y)/p)}{\log p}\Big) - \omega\Big(\frac{\log(X/p)}{\log p}\Big)\Big], \tag{4.1}$$

$$M_{2,2} := \sum_{\sqrt{X}<p\le\sqrt{X+Y}} \frac{X+Y}{p\,\log p}\left[\omega\Big(\frac{\log((X+Y)/p)}{\log p}\Big) - \omega\Big(\frac{\log(X/p)}{\log p}\Big)\right]. \tag{4.2}$$

For $p \le \sqrt{X}$ we have $u_{j,p} \in [1,3]$; by Lemmas 3.2 and 3.3,

$$|\omega(u_{2,p}) - \omega(u_{1,p})| \;\le\; \frac{\log(1+Y/X)}{\log p} \;\ll\; \frac{Y}{X\log p},$$

and hence

$$M_{2,1} \;\ll\; (X{+}Y)\frac{Y}{X}\sum_{y<p\le\sqrt{X}} \frac{1}{p\,\log^2 p} \;\ll\; \frac{Y}{\log^2 y}.$$

On the "very short range" $\sqrt{X} < p \le \sqrt{X+Y}$ the crossing $u = 1$ may occur (due to the extension $\omega(u) = 0$ for $u < 1$); thus we only use $0 \le \omega(u_{2,p}) - \omega(u_{1,p}) \le 1$. Therefore,

$$M_{2,2} \;\le\; \sum_{\sqrt{X}<p\le\sqrt{X+Y}} \frac{X+Y}{p\,\log p}.$$

By the explicit inequality for the prime-counting function, $\pi(b) - \pi(a) \ll \dfrac{b-a}{\log a}$ (with $a = \sqrt{X}$; cf. *Dusart, **Thm. 6.9***; see also *Rosser–Schoenfeld*), since $p \ge \sqrt{X} \ge y$ and thus $\log p \ge \log y$,

$$\sum_{\sqrt{X}<p\le\sqrt{X+Y}} \frac{1}{p\,\log p} \;\le\; \frac{1}{\sqrt{X}\,\log y}\,(\pi(\sqrt{X+Y}){-}\pi(\sqrt{X})) \;\ll\; \frac{1}{\sqrt{X}\,\log y}\cdot\frac{\sqrt{X+Y}-\sqrt{X}}{\log y} \;\ll\; \frac{Y}{X\,\log^2 y}.$$

Hence,

$$M_{2,2} \;\ll\; (X{+}Y)\cdot\frac{Y}{X\,\log^2 y} \;\ll\; \frac{Y}{\log^2 y}.$$

Finally, by Lemma A.3 (applied with $s = X/p$ and $\Delta = Y/p$),

$$R \;\ll\; \sum_{y<p\le\sqrt{X+Y}} \frac{Y/p}{\log^2 p} \;\ll\; \frac{Y}{\log^2 y}.$$

Summing up yields the claimed identity. □

Remark 4.2 (Use in the local proof). When employing Lemma 4.1 inside Lemma 5.1, we keep the decomposition

$$\sum_{y<p\le\sqrt{X+Y}} \Delta_p \;=\; M_1 + M_2 + R \;=\; \frac{Y}{\log y}\,\omega(u) \;+\; M_2 \;+\; O\Big(\frac{Y}{\log^2 y}\Big),$$

where the $O(\cdot)$ term aggregates *only* the error from M_1 (Abel+Mertens) and R. The sum M_2 remains explicit to be combined with $\pi(X{+}Y) - \pi(X)$ in Part (II) of Lemma 5.1.

Boundary lemma for $X = y^2$

Lemma 4.3 (Short increment at the boundary $X = y^2$)**.** *If $y \geq U_0$ and Y satisfies $y \leq Y \leq 2y$, then*

$$\Phi(y^2+Y, y) - \Phi(y^2, y) \;=\; \frac{Y}{2\log y} \;+\; O\Big(\frac{Y}{\log^2 y}\Big).$$

Proof. By Lemma 3.1, with $x \in \{y^2, y^2+Y\}$ and $u = \frac{\log x}{\log y} \in [2, 2+\frac{\log(1+Y/y^2)}{\log y}] \subset [2,3]$,

$$\Phi(x,y) = \frac{x}{\log y}\,\omega\Big(\frac{\log x}{\log y}\Big) + E(x,y), \qquad |E(x,y)| \leq C_0\,\frac{x}{\log^2 y}.$$

Subtracting $x = y^2$ from $x = y^2+Y$ and using $\omega(2) = \frac{1}{2}$ and the variation control on $[2,3]$ (Lemma 3.3),

$$\frac{y^2+Y}{\log y}\big(\omega(2{+}\delta) - \omega(2)\big) \;=\; O\Big(\frac{y^2}{\log y}\cdot\frac{\delta}{1}\Big) \;=\; O\Big(\frac{y^2}{\log y}\cdot\frac{Y}{y^2\log y}\Big) \;=\; O\Big(\frac{Y}{\log^2 y}\Big),$$

where $\delta = \frac{\log(1+Y/y^2)}{\log y} \ll \frac{Y}{y^2\log y}$. Since $Y \leq 2y$, in particular

$$\delta \;\leq\; \frac{Y/y^2}{\log y} \;\leq\; \frac{2}{y\log y}.$$

By Lemma A.2, we have $|E(y^2{+}Y, y) - E(y^2,y)| \;\leq\; C_E\,Y/\log^2 y$ with $C_E \leq 3$. This gives the claimed identity. $\square$

5 Local lemma with uniform error and explicit constants

Lemma 5.1 (Local). *There exist absolute constants* $C_1, C_2 > 0$ *such that, for* $y \geq U_0$, $X \in [y^2, y^3]$ *and* Y *with* $y \leq Y \leq 2y$, *writing* $u := \frac{\log X}{\log y} \in [2,3]$,

$$\#\{X < n \leq X+Y : \; P^{-}(n) > y\} \;\geq\; \frac{Y}{\log y}\,\omega(u) \;-\; C_1\,\frac{Y}{\log^2 y} \;-\; C_2,$$

with

$$C_1 \;=\; c_{\mathrm{inc}} + c_{\mathrm{var}}, \quad c_{\mathrm{inc}} \leq 2, \quad c_{\mathrm{var}} \leq 2.4 \;\; (\textit{for } y \geq U_0), \qquad C_1 \leq 4.4, \qquad C_2 \;=\; 100 \; (\textit{absolute constant}).$$

Detailed proof (short interval via Buchstab — finite-difference version). Equivalently, one can see this by subtracting the form $\Phi(x,y) = \Phi(x,X) + \sum_{y<p\leq X} \Phi(x/p,p)$ at $x = X$ and $x = X+Y$, which produces directly $S = \sum_{y<p\leq X} \Delta_p + \sum_{X<p\leq X+Y} \Phi((X+Y)/p,p)$. We use this to isolate the main terms and residues.

By Buchstab's identity (H–R, Ch. III §6),

$$S := \Phi(X+Y,y) - \Phi(X,y) = \sum_{y<p\leq X} \Delta_p \;+\; \sum_{X<p\leq X+Y} \Phi((X+Y)/p,p),$$

where $\Delta_p := \Phi((X+Y)/p,p) - \Phi(X/p,p)$.

(I) Treating $\sum_{y<p\leq X} \Delta_p$. Slice the primes into

$$\mathcal{P}_1 := \{\, y < p \leq \sqrt{X}\,\}, \qquad \mathcal{P}_2 := \{\, \sqrt{X} < p \leq \sqrt{X+Y}\,\}, \qquad \mathcal{P}_3 := \{\, \sqrt{X+Y} < p \leq X\,\}.$$

For $p \in \mathcal{P}_1$ we have $u_{j,p} = \frac{\log(x_j/p)}{\log p} \in [1,3]$, so Lemma 3.1 applies with $u \in [1,3]$; for $p \in \mathcal{P}_2$ the range is very short and $\Delta_p \geq 0$; for $p \in \mathcal{P}_3$ one has $X/p < p$ and $(X+Y)/p < p$, hence $\Phi(\cdot,p) = 1$ and $\Delta_p = 0$. If $X = y^2$, there are no p in $\mathcal{P}_1$ and we use Lemma 4.3 directly:

$$S = \Phi(X+Y,y) - \Phi(X,y) = \frac{Y}{2\log y} + O\Big(\frac{Y}{\log^2 y}\Big),$$

so the statement follows with $u = 2$.

In the case $X > y^2$, note that $\Delta_p = 0$ for $p > \sqrt{X+Y}$ (since $(X+Y)/p < p$ and also $X/p < p$), so

$$\sum_{y<p\leq X} \Delta_p \;=\; \sum_{y<p\leq\sqrt{X+Y}} \Delta_p.$$

This captures exactly $\mathcal{P}_1 \cup \mathcal{P}_2$ from the slicing above; $\mathcal{P}_3$ does not contribute ($\Delta_p = 0$). Using the decomposition $M_1 + M_2 + R$ from Lemma 4.1, we have uniformly

$$\sum_{y<p\leq\sqrt{X+Y}} \Delta_p \;=\; M_1 + M_2 + R \;=\; \frac{Y}{\log y}\,\omega(u) \;+\; M_2 \;+\; O\Big(\frac{Y}{\log^2 y}\Big),$$

where the $O(\cdot)$ term above gathers *only* the error from M_1 (via Abel+explicit Mertens) and the residue R, keeping M_2 explicit for Part (II). Absorbing the implicit constant in $O(\cdot)$ into c_{inc}, we obtain

$$\sum_{y<p\leq\sqrt{X+Y}} \Delta_p \;=\; \frac{Y}{\log y}\,\omega(u) \;+\; M_2 \;+\; O\Big(\frac{Y}{\log^2 y}\Big), \qquad |O(\cdot)| \;\leq\; c_{\mathrm{inc}}\,\frac{Y}{\log^2 y}, \quad c_{\mathrm{inc}} \leq 2 \;\; (y \geq U_0).$$

For later reference, **defining** $u_{j,p} := \frac{\log(x_j/p)}{\log p}$ with $x_1 = X$ and $x_2 = X{+}Y$, and using the variation control of ω on $[1, 3]$ (Lemmas 3.2, 3.3), we get

$$|\omega(u_{2,p}) - \omega(u_{1,p})| \ \le\ \frac{\log(1 + Y/X)}{\log p} \ \le\ \frac{Y}{X \log p},$$

hence

$$\Big|\frac{X + Y}{p \log p}\, [\omega(u_{2,p}) - \omega(u_{1,p})]\Big| \ \ll\ \frac{Y/p}{\log^2 p}.$$

(II) Treating $\sum_{X<p\le X+Y} \Phi((X{+}Y)/p, p)$ and a lower bound. For $X < p \le X + Y$, we have $1 \le (X{+}Y)/p < 2$; hence $\Phi((X{+}Y)/p, p) = 1$. Thus the sum is $\pi(X{+}Y) - \pi(X)$. As we seek a *lower bound*, it suffices to note $\pi(X{+}Y) - \pi(X) \ge 0$; we will combine it with M_2 next.

For the variation term, now with the correct factor $1/p$, by Lemmas 3.2 and 3.3 we have uniformly $\omega(u_{2,p}) - \omega(u_{1,p}) \ge -\dfrac{\log(1 + Y/X)}{\log p}$. Hence,

$$\sum_{y<p\le X} \frac{X + Y}{p\, \log p}\big[\omega(u_{2,p}) - \omega(u_{1,p})\big] \ \ge\ -(X{+}Y)\, \log(1{+}Y/X) \sum_{y<p\le X} \frac{1}{p\, \log^2 p} \ \ge\ -c_{\mathrm{var}}\, \frac{Y}{\log^2 y},$$

where we used $\log(1 + Y/X) \le Y/X$ and, for $y \le p \le X \le y^3$,

$$\sum_{y<p\le X} \frac{1}{p\, \log^2 p} \ \le\ \frac{1}{\log^2 y} \sum_{y<p\le X} \frac{1}{p} \ =\ \frac{1}{\log^2 y}\Big(\log\log X - \log\log y + O(1)\Big) \ \le\ \frac{\log 3 + O(1/\log y)}{\log^2 y}.$$

(Explicit Mertens; Rosser–Schoenfeld, Thm. 20; Dusart, Thm. 6.10.)

Taking $c_{\mathrm{var}} \le 2.4$ (for $y \ge U_0$) suffices for what follows.

Therefore, since $\sum_{y<p\le\sqrt{X+Y}} \frac{1}{p\, \log^2 p}$ is $\le$ the sum over $y < p \le X$, we also obtain

$$\sum_{X<p\le X+Y} \Phi((X{+}Y)/p, p) \ +\ M_2 \ \ge\ -c_{\mathrm{var}}\, \frac{Y}{\log^2 y}.$$

This allows us to combine M_2 from (I) directly with $\pi(X{+}Y) - \pi(X)$ without resorting to delicate cancellations.

Putting together (I) and (II), and noting $Y \in [y, 2y]$ (no accumulation of $O(1)$ terms), we get

$$S \ \ge\ \frac{Y}{\log y}\, \omega(u) \ -\ (c_{\mathrm{inc}} + c_{\mathrm{var}})\, \frac{Y}{\log^2 y} \ -\ C_2,$$

with absolute C_2 (we take 100). This proves the claim. □

Remark 5.2 (Origin of the term C_2). The absolute term C_2 absorbs boundary contributions that do not scale with $Y/\log^2 y$ and arise in the sum–integral passage and in gluing the cases $X = y^2$ and $X > y^2$: (*i*) discretization on the very short window $\sqrt{X} < p \le \sqrt{X + Y}$ (where we only use trivial majorants), (*ii*) truncations after Abel+Mertens while keeping smooth weights in p, and (*iii*) the "glue" with the boundary Lemma $X = y^2$ (Lemma 4.3). For $y \ge U_0$ all these pieces admit absolute bounds independent of X, Y; for convenience we take $C_2 = 100$, which is largely dominated by $\frac{Y}{\log^2 y}$ when $Y \in [y, 2y]$.

Remark 5.3 (On step (II) of Lemma 5.1). In the term

$$M_2 := \sum_{y<p\le\sqrt{X+Y}} \frac{X+Y}{p\,\log p}\left[\omega\Big(\tfrac{\log((X+Y)/p)}{\log p}\Big) - \omega\Big(\tfrac{\log(X/p)}{\log p}\Big)\right],$$

the contributions with $p > \sqrt{X+Y}$ are identically zero: indeed, $(X+Y)/p < p$ and $X/p < p$, hence $u_{2,p}, u_{1,p} < 1$ and, by the convention $\omega(u) = 0$ on $(0,1)$, we have $\omega(u_{2,p}) - \omega(u_{1,p}) = 0$. Consequently,

$$\sum_{y<p\le X} \frac{X+Y}{p\,\log p}\big[\omega(u_{2,p}) - \omega(u_{1,p})\big] \;=\; \sum_{y<p\le\sqrt{X+Y}} \frac{X+Y}{p\,\log p}\big[\omega(u_{2,p}) - \omega(u_{1,p})\big] \;=\; M_2.$$

Therefore, combining this identity with $\pi(X+Y) - \pi(X) \ge 0$ and the lower bound on the variation derived above, we obtain

$$\sum_{X<p\le X+Y} \Phi((X+Y)/p, p) \;+\; M_2 \;\ge\; -\,c_{\mathrm{var}}\,\frac{Y}{\log^2 y},$$

as claimed.

6 Main theorem

Theorem 6.1. *For every $m \geq 10^6$ and every $x \in \{0, 1, \dots, 8\}$, the block*

$$K_x = \{\, m^2 + xm + 1, \dots, m^2 + (x+1)m \,\}$$

contains at least one m-rough integer.

Proof. Fix x and set $X = m^2 + xm$, $Y = y = m$. Then

$$u = \frac{\log X}{\log m} = 2 + \delta, \qquad \delta = \frac{\log(1 + x/m)}{\log m} \in \Big(0, \frac{9}{m \log m}\Big).$$

By Lemma 3.3, $\omega(u) \geq \frac{1}{2} - \delta/4$; in particular, for $m \geq 10^6$, $\omega(u) \geq \frac{1}{2} - \frac{9}{4m \log m}$. By Lemma 5.1 with $C_1 \leq 4.4$ and $C_2 = 100$,

$$\#\{n \in K_x :\ P^-(n) > m\} \ \geq\ \frac{m}{\log m}\Big(\frac{1}{2} - \frac{\delta}{4}\Big) \ -\ 4.4\,\frac{m}{\log^2 m} \ -\ 100.$$

For $m = 10^6$, $\frac{m}{2 \log m} \approx 3.62 \cdot 10^4$ and $4.4\,\frac{m}{\log^2 m} \approx 2.30 \cdot 10^4$; while the term $\frac{m}{\log m}\frac{\delta}{4} \leq \frac{9}{4 \log^2 m} \approx 0.01$ is negligible. Thus the margin is $\gtrsim 1.3 \cdot 10^4 \gg 1$, and each K_x contains at least one m-rough integer. $\square$

Remark 6.2 (Explicit range with the current constants). With $C_1 \leq 4.4$ and $C_2 = 100$ from Lemma 5.1, numerical calculations (reproducible artifact) indicate that

$$\frac{m}{\log m}\left(\tfrac{1}{2} - \frac{9}{4m \log m}\right) \ -\ 4.4\,\frac{m}{\log^2 m} \ -\ 100 \ \geq\ 1$$

already holds for all $m \geq 18{,}794$ and fails narrowly at $m = 18{,}793$. This observation is not used in the formal proof of Theorem 6.1, where we adopt the conservative junction $m \geq 10^6$.

Remark 6.3 (Compatibility with the margin at $m = 10^6$). There is no conflict between Remark 6.2 and the analytic bound used for $m \geq 10^6$. The remark merely identifies an effective threshold (with the current constants) from which the right-hand side of the lower bound is ≥ 1. At $m = 10^6$, the same inequality yields substantial slack:

$$\frac{m}{\log m}\left(\tfrac{1}{2} - \frac{9}{4m \log m}\right) \ -\ 4.4\,\frac{m}{\log^2 m} \ -\ 100 \ \geq\ 24050 \ -\ 23105 \ -\ 100 \ =\ 845.$$

Thus, the case $m \geq 10^6$ features a margin well above 1. We keep the theorem stated for $m \geq 10^6$ out of conservatism; the remark simply suggests that the effective threshold in m can be lowered.

7 Final remarks

We performed a computational sweep up to 10^6 covering the nine blocks $K_0, \ldots, K_8$ (reproducible artifact; see Remark 7.1), which *integrates the proof* on the range $2 \le m < 10^6$. For $m \ge 10^6$, the local lemma with effective constants ($C_0 \le 2$, $C_1 \le 4.4$, $C_2 = 100$) guarantees a comfortably positive margin in each block; numerical calculations suggest positivity well below this threshold, but we keep $m \ge 10^6$ as the gluing boundary, for conservatism.

To broaden the scope, one may fix m and sweep additional blocks and/or increase m to enlarge the theoretical margin. Pushing m to 10^{10} is unnecessary here and would render the numerical exploration costly; the mechanism is recorded for future work.

Remark 7.1 (Computational artifacts and formal integration).

Finite range $2 \le m < 10^6$**.**
Verified by a deterministic sieve sweep: for each m and $x = 0, \ldots, 8$ we find an m-rough witness in every K_x. Code, logs, and reproduction instructions are available as a supplementary artifact on Zenodo (doi.org/10.5281/zenodo.17137291). In the Lean formalization, this range is encapsulated by a single axiom `RoughBlocks.Light.budget_verified_by_scan`.

Asymptotic range $m \ge 10^6$**.**
The bridge inequality comparing the analytic lower bound $LB(m, x)$ with the actual count in the block K_x is validated *uniformly* by a Julia certificate with interval arithmetic (256 bits), generated by `external/julia/uniform_bridge_certificate.jl` and stored as `certs/uniform-bridge-ge-1e6.json` in the code repository (github.com/mariltondev/rough-blocks-lean). In the formalization, this is referenced as the axiom `RoughBlocks.External.bridge_ge_1e6_uniform`.

Appendices

A Proof that $C_0 \leq 2$ in the global lemma

In this appendix we give an *explicit proof* that $|E(x,y)| \leq C_0\, x/\log^2 y$ with $C_0 \leq 2$ for $y \geq U_0 = 10^3$, anchoring the effective forms used. We use Buchstab's identity *in summatory form* (H–R, Ch. III §6):

$$\Phi(x,y) = 1 + \sum_{y<p\leq x} \Phi\Big(\frac{x}{p}, p\Big).$$

The passage to integral forms uses explicit Mertens with controlled error (below); then we follow the standard scheme: main term via $\omega(u)$ and the error as the sum of:

(i) Mertens error (sums over primes)

We use the effective form (for $t \geq y \geq 10^3$)

$$\sum_{p\leq t} \frac{1}{p} = \log\log t + B_1 + R_1(t), \qquad |R_1(t)| \leq \frac{A}{\log t} + \frac{B}{\log^2 t} \quad (t \geq 10^3),$$

with explicit A, B (cf. *Rosser–Schoenfeld, **Thm. 20**, Eq. (4.7)*; and *Dusart, **Thm. 6.10***). For $t \geq 10^3$ one can comfortably take $A \leq 1$, $B \leq 1$. By integration by parts in the standard sieve transforms, we get

$$|E_A(x,y)| \;\leq\; c_1 \frac{x}{\log^2 y}, \qquad c_1 \leq 1.$$

(ii) Variation of ω on $[1,3]$

On $[1,2]$, $\omega(u) = 1/u$ gives exact control; on $[2,3]$, by *Lemma 3.3*,

$$|\omega(u+\delta) - \omega(u)| \leq \frac{\delta}{4}.$$

Integrating the variation terms arising after the sum–integral switch (H–R, Ch. III §6), we get

$$|E_B(x,y)| \;\leq\; c_2 \frac{x}{\log^2 y}, \qquad c_2 \leq 1.$$

Adding (i) and (ii),

$$|E(x,y)| \;\leq\; (c_1 + c_2) \frac{x}{\log^2 y} \;\leq\; 2 \frac{x}{\log^2 y},$$

hence $C_0 \leq 2$ for $y \geq 10^3$, as required.

Remark: any improvement in $R_1(t)$ (explicit Mertens) or in the variation control on $[2,3]$ reduces C_0; $C_0 \leq 2$ suffices for the Theorem.

In particular, for $2 \leq u \leq 3$ and $\delta \geq 0$, by Lemma 3.3 we have $\sup_{2\leq u\leq 3} |\omega'(u)| \leq \frac{1}{4}$, hence $|\omega(u+\delta) - \omega(u)| \leq \delta/4$. Thus there is no need to appeal to additional integrals nor does the Euler–Mascheroni constant intervene.

Remark A.1 (Use of the short-error lemmas). For the reader's orientation: we use Lemma A.2 only in the case $X = y^2$ (i.e., in the application of Lemma 4.3), and Lemma A.3 for the control of the term R in Lemma 4.1 (applied with $s = X/p$ and $\Delta = Y/p$).

Short increment of the error at the edge $X \asymp y^2$

Lemma A.2 (Short increment of the error)**.** *If $y \geq U_0$, $x \in [y^2, y^2+2y]$ and $y \leq \Delta \leq 2y$, then*

$$\big|E(x+\Delta, y) - E(x,y)\big| \;\leq\; C_E\,\frac{\Delta}{\log^2 y},$$

with an absolute effective constant $C_E \leq 3$.

Sketch with constants. By the summatory form of Buchstab, for $t \in \{x, x+\Delta\}$: $\Phi(t,y) = 1 + \sum_{y<p\leq t}\Phi(t/p,p)$. Subtracting, we isolate the increment of the main term and of the residue: $|E(x{+}\Delta,y) - E(x,y)| \;\leq\; \sum_{y<p\leq\sqrt{x+\Delta}}\frac{\Delta/p}{\log^2 p} \;+\; \mathcal{V}$, where $\mathcal{V}$ comes from the variation of ω on $u \in [1,3]$. Applying Abel with $A(t) = \sum_{y<p\leq t} 1/p$ and explicit Mertens $A(t) = \log\log t + B_1 + R_1(t)$, $|R_1(t)| \leq A/\log t + B/\log^2 t$ (Rosser–Schoenfeld, Thm. 20; Dusart, Thm. 6.10), we get

$$\sum_{y<p\leq\sqrt{x+\Delta}}\frac{\Delta/p}{\log^2 p} \;\leq\; \frac{\Delta}{\log^2 y}\Big(1+O\big(\tfrac{1}{\log y}\big)\Big).$$

Meanwhile the variation of ω on $[1,3]$ is $\ll \Delta/(p\log^2 p)$ (Lemmas 3.2, 3.3) and sums to the same order. For $y \geq 10^3$ we may take $A \leq 1$, $B \leq 1$, which gives comfortably $C_E \leq 3$. □

General short increment of the error

Lemma A.3 (General short increment of the error)**.** *If $y \geq U_0$, $p \geq y$, $s \in [1,p^2]$ and $0 \leq \Delta \leq 2$, then*

$$\big|E(s+\Delta, p) - E(s,p)\big| \;\leq\; C_E\,\frac{\Delta}{\log^2 p},$$

with an absolute effective constant $C_E \leq 3$.

Sketch with constants. From Buchstab's summatory identity, for $t = ps$ and $t = p(s+\Delta)$:

$$\Phi(t,p) = 1 + \sum_{p<q\leq t}\Phi(t/q,q).$$

Subtracting, the increment of the residue is $|E(s{+}\Delta,p) - E(s,p)| \;\leq\; S_1 + S_2$, where S_1 comes from the $1/q$ weight after Abel and S_2 from the variation of ω. For S_1, using Abel with $A(u) = \sum_{p<q\leq u} 1/q$ and explicit Mertens $A(u) = \log\log u - \log\log p + O(1/\log p)$,

$$S_1 \;\ll\; \Delta \sum_{p<q\leq\sqrt{p(s+\Delta)}}\frac{1}{q\,\log^2 q} \;\leq\; \Delta\int_p^\infty \frac{du}{u\,\log^2 u} \;=\; \frac{\Delta}{\log p}\Big(1+O\big(\tfrac{1}{\log p}\big)\Big).$$

The extra $1/\log p$ factor comes from the smooth weight of the main term, hence $S_1 \ll \Delta/\log^2 p$. For S_2, on $u \in [1,3]$ we have $|\omega(u+\varepsilon) - \omega(u)| \ll \varepsilon$ on $[1,2]$ and $\ll \varepsilon/4$ on $[2,3]$ (Lemmas 3.2, 3.3), which yields

$$S_2 \;\ll\; \Delta \sum_{p<q\leq\sqrt{p(s+\Delta)}}\frac{1}{q\,\log^2 q} \;\ll\; \frac{\Delta}{\log^2 p}.$$

Taking $A, B \leq 1$ in the explicit Mertens forms (Rosser–Schoenfeld, Thm. 20; Dusart, Thm. 6.10), we obtain the effective constant $C_E \leq 3$. □

Technical note (finite differences vs. derivatives)

In all local passages we avoid derivatives of E. We use only: (i) Buchstab's identity for Φ on short increments Δ, (ii) the variation control of ω on $[2, 3]$ (Lemma 3.3), and (iii) partial sums + explicit Mertens to convert the sums over p into the main terms and residues $\ll \Delta/\log^2 y$. This guarantees uniformity in $X \in [y^2, y^3]$, $Y \in [y, 2y]$ without additional regularity hypotheses on E.

B Mapping between the paper and the Lean formalization

B.1 Project repository (GitHub)

`github.com/mariltondev/rough-blocks-lean`

Modules currently in the repository

- `RoughBlocks/Defs.lean` — basic definitions:
  `mRough`, `K`, `BlockHasMRough`,
  `roughSetInBlock`, `countRoughInBlock`,
  existential lemma `exists_of_count_ge_one`.
- `RoughBlocks/Check.lean` — API sanity (sentinel examples).
- `RoughBlocks/Scratch.lean` — demo executable (target `roughblocks_scratch`).
- `RoughBlocks/Heavy/Buchstab/Core.lean` — combinatorial primitives:
  `isRoughGT`, `isRoughGE`, `PhiGT`, `PhiGE`, `primesIn`,
  decidable instances and basic lemmas used in the identities.
- `RoughBlocks/Heavy/Buchstab/MinFac.lean` — thin layer over `Nat.minFac`:
  lemmas `minFac_prime_of_two_le`, `minFac_le_of_prime_dvd`, `minFac_dvd_eq`.
- `RoughBlocks/Heavy/Buchstab.lean` — (optional) aggregator that re-exports the Buchstab core.
- `RoughBlocks/Heavy/WindowLink/Core.lean` — "window $(N_1, N_2] \leftrightarrow \Phi$-difference":
  `PhiGE_diff_window_eq`, `PhiGE_diff_window_ge`, `PhiGE_diff_window_ge_div`,
  `sum_card_eq_sum_diff_phiGE`.
- `RoughBlocks/Heavy/WindowLink/Block.lean` — block $\leftrightarrow$ window bridge (case $p := m+1$):
  `countRoughInBlock_as_window`,
  `strict_window_card_eq_phiDiff_succ` and its $\mathbb{R}$ version,
  `countRoughInBlock_eq_phiDiff_succ_real`.
- `RoughBlocks/Heavy/Identity.lean` — Buchstab identities (global and window):
  `buchstab_identity_PhiGT`, `buchstab_identity_window_core`, `countWindow_as_sumDiff_real`,
  version with $1 \le X$; equivalences `mRough`$\leftrightarrow$`isRoughGT` for $n \ge 1$;
  `isRoughGT_iff_isRoughGE_succ_of_two_le`; monotonicity `PhiGE_mono_in_X`.
- `RoughBlocks/Heavy/Numeric.lean` — analytic/conservative layer (no external axioms):
  `LB : ` $\mathbb{N} \to \mathbb{N} \to \mathbb{R}$, `margin : ` $\mathbb{N} \to \mathbb{R}$, `marginR : ` $\mathbb{R} \to \mathbb{R}$,
  `LB_ge_margin'`, milestones `margin_ge_one_at_1e6`, `margin_ge_one_from_1e6`,
  `budget_conservative_formal` ($m \ge 10^6$).
- Auxiliary modules (analytic parts):
  `RoughBlocks/Heavy/Bridge.lean`
  `RoughBlocks/Heavy/Interface.lean`
  `RoughBlocks/Heavy/Log10Bounds.lean`
  `RoughBlocks/Heavy/Omega.lean`

- `RoughBlocks/Light/Concrete.lean` — concrete translations/auxiliaries used in the public statement.
- `RoughBlocks/Light/Existence.lean` — unified existence theorem (Light layer);
  external axiom `budget_verified_by_scan` (range $2 \le m < 10^6,\ 0 \le x \le 8$).
- `RoughBlocks/Light/Export.lean` — public *facade*; re-exports the proven API:
  target statements such as `main_theorem_ge_1e6` and `budget_main`.
- `RoughBlocks/Light/Spec.lean` — audit target (#print axioms);
  consolidates external dependencies visible in the `Light` layer.
- `RoughBlocks/External/BridgeUniform.lean` — external axiom `bridge_ge_1e6_uniform`
  imports the Julia certificate (JSON) and exposes it as a uniform bridge in the formalization.

Note. The only external dependency assumed is the computational check encapsulated by the axiom `RoughBlocks.Light.budget_verified_by_scan`, whose scope is exactly: for $2 \le m < 10^6$ and $0 \le x \le 8$, we have $\mathrm{LB}(m, x) \ge 1$. The corresponding result was verified by a deterministic sieve sweep, iterating $x = 0, \ldots, 8$ for each m and finding an m-rough witness in every block K_x. Code, logs, and reproduction instructions are published as a Zenodo artifact (doi.org/10.5281/zenodo.17137291). All remaining parts of the development—Buchstab identities, window bijections and arithmetic, the block bridge, and the formal range $m \ge 10^6$—are fully proved in Lean, with no additional assumptions beyond the system's classical axioms.

The preprint is available at this link:

https://doi.org/10.5281/zenodo.16734582

SHA256 hash of Version 1 (available at Zenodo)

`39427b6d90cd86895ad31263ddfc3201309fdc582a5e24d7e97cd07c12afbb02`

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